

5.Crypt Arithmetic Problem

Aim

To solve a Crypt Arithmetic puzzle using AI.

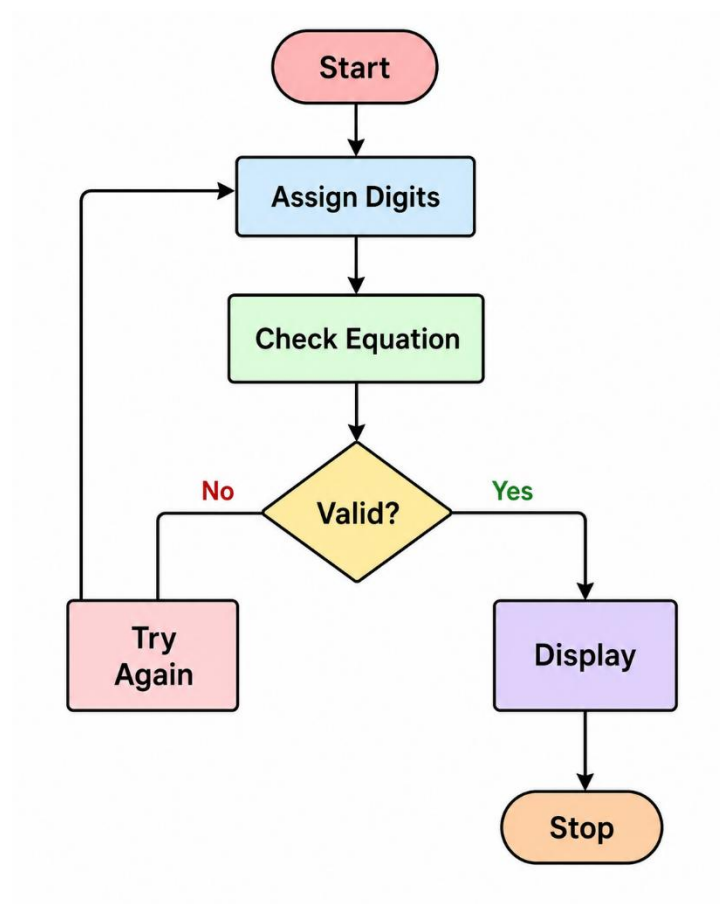
Objective

To assign digits to letters such that arithmetic operations are satisfied.

Algorithm

1. Assign unique digits to letters.
2. Check arithmetic validity.
3. Continue until a valid solution is found.
4. Display solution.

Flowchart:



Python code:

```
from itertools import permutations
```

```
word1 = input("Enter first word: ").upper()
```

```
word2 = input("Enter second word: ").upper()
```

```
result = input("Enter result word: ").upper()
```

```
letters = ''.join(set(word1 + word2 + result))
```

```
if len(letters) > 10:
```

```
    print("Too many unique letters (maximum 10 allowed).")
```

```
else:
```

```
    digits = permutations('0123456789', len(letters))
```

```
    for p in digits:
```

```
        d = dict(zip(letters, p))
```

```
        if d[word1[0]] == '0' or d[word2[0]] == '0' or d[result[0]] == '0':
```

```
            continue
```

```
        num1 = int(''.join(d[ch] for ch in word1))
```

```
        num2 = int(''.join(d[ch] for ch in word2))
```

```
        num3 = int(''.join(d[ch] for ch in result))
```

```
        if num1 + num2 == num3:
```

```
            print("\nSolution Found:")
```

```
            print(num1, "+", num2, "=", num3)
```

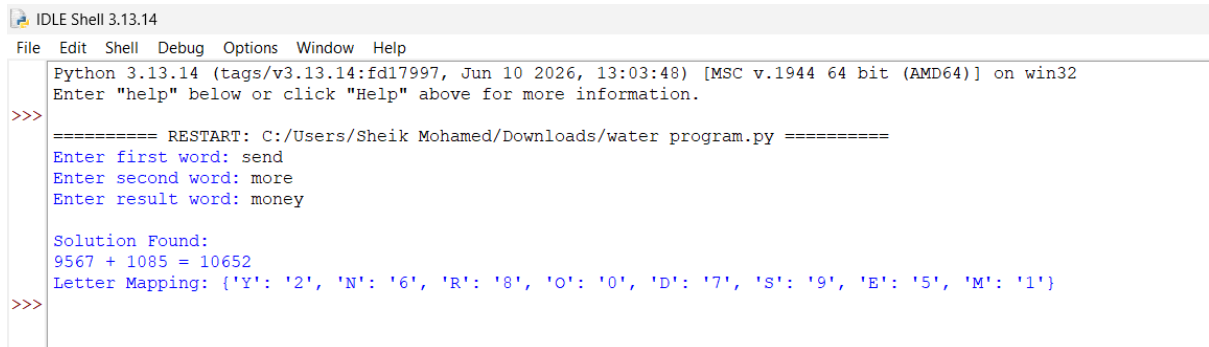
```
            print("Letter Mapping:", d)
```

break

else:

print("No solution found.")

output:



```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sheik Mohamed/Downloads/water program.py =====
Enter first word: send
Enter second word: more
Enter result word: money

Solution Found:
9567 + 1085 = 10652
Letter Mapping: {'Y': '2', 'N': '6', 'R': '8', 'O': '0', 'D': '7', 'S': '9', 'E': '5', 'M': '1'}
>>>
```

Result:

The puzzle was solved successfully.

Conclusion:

Crypt Arithmetic can be solved using constraint satisfaction techniques.